

Eustace the Pirate Monk — Specification

CS 2114 Project 1, Deliverable 2 — Group 87 (Marcos Salas, Aidan McIlvenni)

Java 8, default package, nothing beyond `java.util` and `student.jar`. Numbers marked *default* live in named constants; Marcos tunes them in week 2, and section 6's expected values are recomputed when they change. Attached: `docs/class-diagram.png` (full UML) and `docs/channel-map.png` (the map).

1. Class design

Twelve classes and one interface, each with one job:

1. `Game`: runs the read–dispatch–print loop, computes the current mode, and is the only class that prints. Holds `main`.
2. `CommandParser`: turns one raw line into a `Command`, holds the table of verbs and their usage lines, and owns every typing-slip rule (`DESIGN.md` §4 cases 1–6, 8–11, 25, 27).
3. `Command`: an immutable value, a verb word and its argument string.
4. `ChannelMap`: builds the fixed 18-stop map and finds a stop from the name typed.
5. `Location`: one stop, with its key, name, kind, encounter chance and the stops one move away.
6. `Port`: a `Location` where the ship moors, with its nation and prices.
7. `Ship` (abstract): what both sides share, hull, cannons, armour, gold, rum and a `Crew`, plus the rules for dealing and taking damage.
8. `PlayerShip`: where the player is, notoriety, ports visited, and every purchase.
9. `EnemyShip`: a ship the player meets, built from an `EnemyType` and the player's notoriety.
10. `EnemyType`: the base numbers for one kind of enemy, with three shared instances, `MERCHANT`, `PIRATE` and `COAST_GUARD`, as static final constants.
11. `Crew`: head count, morale and greed, and the rules that move them.
12. `Encounter`: one meeting with an enemy, from the roll that creates it through the fight or the flight to the plunder.
13. `Describable` (interface): `String describe()`, implemented by `Location`, `Ship` and `Encounter`, so `Game` prints any of them the same way.

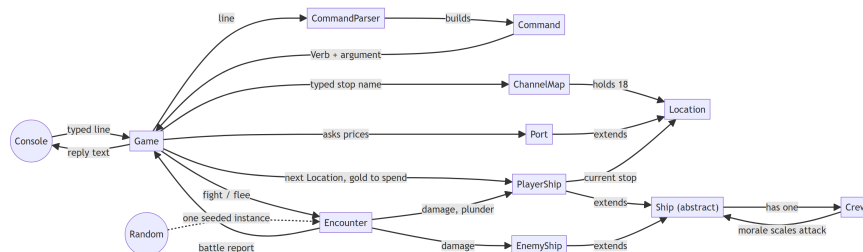
Fixed vocabularies are `String` constants: modes `Game.PORT`, `SEA`, `ENCOUNTER`, `OVER`; kinds `Location.PORT`, `SEA_LANE`, `HIGH_SEAS`; nations "English", "French", "Pirate"; and the verbs `look`, `status`, `sail`, `repair`, `hire`, `buy`, `bonus`, `fight`, `flee`, `retire`, `help`, `quit`.

No class both parses text and changes ship values, and no class both holds game state and prints it.

Why this split: both sides fight by the same rules, so `Encounter` calls `attackStrength()` and `takeDamage()` on a `Ship` without knowing which side it holds. `Crew` is separate because morale scales attack for both sides, and a broken crew makes an enemy surrender or the player's crew mutiny. Cannons and armour are levels 1 to 5, so `buy` takes three fixed words and no port runs out of stock; every port charges the same prices, with `Sark` halving rum. The mode is never stored: `mode()` returns `OVER` after the run ends, `ENCOUNTER` while `encounter != null`, `PORT` at a port, and `SEA` otherwise.

The map: seven ports (three English, three French, and `Sark`, the pirate home), eight sea lanes each joining two ports, and three high-seas stretches touching the nearest ports. One `sail` moves one stop, so `Dover` to `Boulogne` is `sail dover boulogne` then `sail boulogne`; `find` hyphenates the words typed and, failing that, tries the first word alone. Each move into a sea stop makes the crew drink and rolls one encounter.

2. System diagram



Each arrow points the way the labelled thing travels. One turn: `Game` reads a line, `CommandParser` returns a `Command`, `Game` checks the verb against `mode()`, calls the object that owns that verb, and prints what comes back.

3. Data & state

Every field is private, and final where said below. Where a field lists bounds, such as hull from 0 to `maxHull` or morale from 0 to 100, the methods that change it clamp to those bounds, so no caller can drive it outside.

- Game: Scanner in; final ChannelMap map; final PlayerShip player; Encounter encounter, null unless a fight is on; final Random random, the only one in the program; boolean running; static final int GOLD_TARGET = 1000 *default*.
- CommandParser: final HashMap<String, String> usages, verb word to usage line, filled in the constructor with the twelve verbs: one lookup answers both "is this a verb" and "what is its usage". Command: final String verb, the first word in lower case or "" for a blank line; final String argument, lower case, single-spaced, with quotes and ., !? stripped from each word's ends, and "" when absent.
- ChannelMap: final HashMap<String, Location> stops, 18 entries keyed by stop key, fixed once standard() returns; Port home. A HashMap because every sail looks up one typed key in a map built once.
- Location: final String key, name and kind; final int encounterPercent, 0 for a port, 35 for a sea lane and 60 for the high seas *default*; ArrayList<Location> neighbours, 2 to 4 entries, where if A lists B then B lists A, and a stop never lists itself or a duplicate. getNeighbours() hands back the list wrapped in Collections.unmodifiableList, so Game can read and loop over the neighbours while add or remove on what it got throws UnsupportedOperationException; links are made only by connect. The list keeps the order look prints, and as an adjacency list it holds only the 25 links.
- Port adds final String nation and the *default* prices REPAIR_PRICE = 2, HIRE_PRICE = 15, RUM_PRICE = 4 and UPGRADE_PRICE = 100, all gold per unit, where reaching level n costs UPGRADE_PRICE × n.
- Ship: final String name; int hull, 0 to maxHull; final int maxHull; int cannons, 1 to 5; int armour, 0 to 5; int gold and int rum, never negative; final Crew crew. Subclasses change hull, cannons and armour only through protected setters.
- PlayerShip adds Location location, never null; int notoriety; int portsVisited; static final int MAX_LEVEL = 5. It starts *default* with hull 100, crew 20 at morale 70, cannons 1, armour 1, gold 200 and rum 30, moored at Sark.
- EnemyShip adds final EnemyType type. For player notoriety n its constructor sets hull to base + 2n, crew to base + n/2 capped at 40, cannons and armour to base + n/10 each capped at 5, gold to base + 5n, rum to its crew count, and morale to the type's.
- EnemyType: final String name, and final int hull, crew, cannons, armour, gold, morale and gain. The three *default* constants, in that order: MERCHANT 40, 8, 1, 0, 150, 30, 1; PIRATE 60, 15, 2, 1, 100, 60, 2; COAST_GUARD 80, 20, 3, 2, 50, 80, 3.
- Crew: int count, 0 to MAX_COUNT = 40; int morale and int greed, both 0 to 100, where greed starts at 0 and only the player's crew ever gains it. *Default* steps: 5 morale per man lost, 10 when the rum runs out, 10 gained on a win, 15 lost on a flee, and 1 greed per 20 gold plundered.
- Encounter: final PlayerShip player, EnemyShip enemy and Random random; FLEE_DAMAGE = 10 and ROLL_RANGE = 6 *default*.

How the numbers move:

1. moraleFactor() is 0.5 + morale/200, and attackStrength() is (5 × cannons + crew/2) × moraleFactor, floored, at least 1: a fresh player ship at morale 70 gives 12.
2. takeDamage(raw) lands raw – 2 × armour, at least 1 when raw is above 0: hull drops by that much and the crew loses landed/5 men.
3. isDefeated() is true at hull 0, at crew 0, or when crew.isBroken(), which is morale 0 or greed 100.
4. A fight round: the player fires attackStrength() + random.nextInt(ROLL_RANGE); if the enemy still stands it fires back the same way; rounds repeat until one side is defeated. A win moves the enemy's gold and rum across, adds the type's gain to notoriety, and calls crew.onWin(gold).
5. Each sail into a sea stop: the crew needs (count + 9)/10 bottles and drinks them, or drinks what is left and loses 10 morale. Then roll draws nextInt(100) against the stop's chance, and a second draw picks the type: in a sea lane below 70 is a merchant, else the coast guard; on the high seas below 50 is a pirate, else a merchant.
6. A purchase buys the smallest of the amount typed, gold divided by price, and the room left, then charges units × price.

4. Method signatures

"Refuses" means the method returns 0 or false and changes nothing, and dispatch turns that into the section 5 message. IAE means it throws IllegalArgumentException.

Game: Game(Scanner in, ChannelMap map, Random random); static void main(String[] args) builds a game on System.in, ChannelMap.standard() and new Random() and runs it; void run() prints the opening scene and loops read, parse, dispatch, print until mode() is OVER; String dispatch(Command c) applies one command in the current mode and returns the text to print; String mode(); PlayerShip getPlayer(); Encounter getEncounter().

CommandParser: CommandParser() fills the verb table; Command parse(String line) normalises the line and returns a Command holding the first word and the rest, or "" for a blank line; boolean isVerb(String word); String usageOf(String verb) gives the usage line or null; String allUsages() joins every usage line for help; static int parseCount(String text) returns an int of 0 or more, or throws NumberFormatException for anything else.

Command: Command(String verb, String argument), IAE on a null verb; String getVerb(); String getArgument(); String getWord(int i) gives the i-th argument word or ""; String toString() gives "sail over".

ChannelMap: static ChannelMap standard() builds the 18 stops and 25 links; void add(Location stop), IAE on a duplicate key; void connect(String keyA, String keyB) links both ways, IAE on an unknown key; Location find(String name) gives the stop or null; Port getHome().

Location implements Describable: Location(String key, String name, String kind, int encounterPercent); void connect(Location other) adds each to the other once, IAE on itself; boolean isNextTo(Location other); List<Location> getNeighbours(); getters; String describe() gives name, kind and neighbours. Port: Port(String key, String name, String nation) passes kind PORT and chance 0 upward; String getNation(); int rumPrice(), halved at Sark; int upgradePrice(int nextLevel); describe() adds nation and prices.

Ship implements Describable: Ship(String name, int maxHull, int cannons, int armour, Crew crew, int gold, int rum); int attackStrength(); int takeDamage(int raw) returns hull lost; boolean isDefeated(); void addGold(int); void spendGold(int), IAE beyond what is held; void addRum(int); int takeRum(int) returns bottles taken, capped at the hold; Crew getCrew(); a getter per field; abstract String describe().

PlayerShip: PlayerShip(String name, Port home); void moveTo(Location d), IAE unless d is a neighbour, counting a port arrival; int repair(int points, int pricePerPoint), int hire(int men, int pricePerMan) and int buyRum(int bottles, int pricePerBottle) buy up to the amount, capped by gold and room, and return the units bought; boolean upgradeCannons(int price) and boolean upgradeArmour(int price) refuse at MAX_LEVEL or without the gold; boolean payBonus(int gold) refuses beyond what is held, else hands the gold to the crew; void addNotoriety(int); getters; describe() is the status report.

EnemyShip: EnemyShip(EnemyType type, int notoriety), IAE on null or a negative; EnemyType getType(); describe() is what the lookout sees. EnemyType: EnemyType(String name, int hull, int crew, int cannons, int armour, int gold, int morale, int gain) and a getter per field; the three constants are the only instances the game makes.

Crew: Crew(int count, int morale) caps both; int lose(int men) and int hire(int men) return the number removed or added after capping; double moraleFactor(); int drink(int bottlesAvailable) returns bottles drunk and drops morale when short; void onWin(int plunder); void onFlee(); void receiveBonus(int gold) lowers greed by the gold per head and raises morale by half of that; boolean isBroken(); getters.

Encounter implements Describable: Encounter(PlayerShip player, EnemyShip enemy, Random random); static Encounter roll(Location where, PlayerShip player, Random random) gives a new encounter or null; String fight() resolves the whole battle and returns the narration; String flee() applies FLEE_DAMAGE and onFlee(); EnemyShip getEnemy(); String describe().

5. Where validation lives

Numbers are DESIGN.md §4 cases; every response leaves all fields unchanged unless the line says otherwise. dispatch checks the mode first: fight and flee need ENCOUNTER; repair, hire, buy and bonus need PORT; sail is refused in ENCOUNTER.

1. Blank line: parse gives verb "", dispatch returns "", and the prompt comes back.
2. Unknown verb: isVerb is false, and dispatch prints "I don't understand 'sail'. Type help."
3. Missing argument: dispatch sees an empty argument and prints that verb's usage line.
4. Extra words: find falls back to the first word and amounts read getWord(0), so sail dover now please echoes "sail dover" and proceeds. 5, 6, 25. Case, spacing, quotes, punctuation and invisible characters: parse strips, lower-cases, splits on \s+ and trims " ' . , ! ? from each word's ends.
5. A real stop that is not adjacent: find succeeds but isNextTo is false, so dispatch lists the neighbours.
6. sail help: the verb is the first word only, find("help") gives null, and the reply is "no such place". 9, 11. A negative amount, "ten" or "10.1": parseCount throws NumberFormatException, dispatch catches it and asks for a whole number of 0 or more.
7. Zero: parseCount gives 0, the purchase returns 0, and the reply is "bought 0".
8. Overflow: PlayerShip divides gold by price before multiplying, so the cost never passes the gold held and hire 2147483647 buys what 200 gold covers.
9. More than the player can afford or hold: the purchases cap by gold, maxHull and MAX_COUNT and report the units bought, the one case that does change state. 14, 15. A verb in the wrong mode: the dispatch mode check prints "nothing to fight" or "you must be in port".
10. An unknown item: dispatch(BUY) sees a word outside cannons, armour and rum, and lists those three.
11. retire away from home or short of gold: dispatch compares the stop with home and the gold with GOLD_TARGET, and prints the gold still needed.
12. The quit confirmation: dispatch reads one more line and ends the run only on "y" or "yes"; anything else returns to the prompt. 19, 20. Spamming look or status, and sail <current stop>: those branches never call roll or drink, and dispatch(SAIL) refuses a target equal to the current stop with "you are already there".
13. Repeated flee: every flee applies FLEE_DAMAGE, so the hull drains until isDefeated ends the run as a sinking.

14. An upgrade at the maximum: upgradeCannons and upgradeArmour check the level before the gold, so the gold is untouched. 23, 27. A 100,000-character line, and injection: isVerb looks the first word up in the fixed verb table, and find and buy compare against fixed keys and three item words, so the line is refused as unknown and typed text is only ever data.
15. Input ends: run checks hasNextLine() before every read, prints "input ended" and sets running to false.
16. Accents and emoji: find gives null and no item word matches, so the reply is the case 8 or case 16 message.

6. Test plan

One student. TestCase class per production class, in the default package so tests reach the protected setters. FixedRandom is a test-only Random whose nextInt returns a set value: 0 forces an encounter, the first-listed type and a zero damage roll; 99 forces none. A *fresh* player is a new PlayerShip at Sark. Parameterless methods get an edge state in place of bad input, getters are covered by their constructor's test, and bracketed numbers are DESIGN.md §4 cases, all 27 of which appear.

Game:

- Game(...) and run: a fresh game reports mode PORT at Sark, and the script look, status, quit, yes prints "Sark" and "Hull 100/100" and ends at OVER. A null map throws IAE, and a script of a blank line, sial, then end of input prints the unknown-command text and "input ended" without an exception [1, 2, 24].
- dispatch, normal: sail barfleur sark reaches the lane with rum 28; hire 5 gives crew 25 and gold 125; with FixedRandom(0), fight at the lane ends with gold 350 and no encounter; retire with 1000 gold ends at OVER; look, status and help print the stop name, "Morale 70" and every usage line. Bad: bare sail gives the usage line [3]; sail dover names Barfleur–Sark and West Channel [7]; sail sark says the ship is already there [20]; hire 2147483647 buys 13 men for 195 gold [12, 13]; hire at sea says the ship must be in port [15]; buy spyglass lists the three items [16]; buy cannons at level 5 keeps gold 200 [22]; fight in port says nothing to fight [14]; flee after setHull(5) sinks the ship [21]; retire with 400 gold gets "600 more needed" [17]; yse at the quit question keeps the run alive [18]; three looks at sea leave the encounter null [19]; help sail ignores the extra word [4].

CommandParser and Command:

- parse, isVerb, usageOf, allUsages and parseCount: SAIL "Dover". gives "sail" and "dover" [5, 6, 25]; usageOf("sail") gives "sail" and allUsages() holds all twelve; "25" gives 25 and "0" gives 0 [10]. Tabs and spaces give verb "" [1]; sial and System.exit(0) fail isVerb, and usageOf gives null for them [2, 27]; "-50", "ten", "10.1" and "9999999999" throw NumberFormatException [9, 11].
- Command(...), getWord and toString: from ("buy", "rum 20"), getWord(1) gives "20" and toString gives "buy rum 20"; a null verb throws IAE and getWord(5) gives "".

ChannelMap, Location and Port:

- standard, add, connect and find: 18 stops, home sark, every link recorded both ways; dover boulogne finds the lane and dover now please finds Dover [4]. A duplicate key and connect("dover", "atlantis") throw IAE; "atlantis", "help" and "dövr" give null [8, 26].
- Location(...), connect, isNextTo, getNeighbours and describe: a lane built with chance 35 reports it; after a.connect(b), a.isNextTo(b) is true; Sark lists Barfleur–Sark then West Channel and its description names both. a.connect(a) throws IAE, connecting a pair twice lists each once, and add on the returned list throws UnsupportedOperationException.
- Port(...), getNation, rumPrice, upgradePrice and describe: Dover reports "English", chance 0, rum 4 and upgradePrice(2) of 200, and its description holds the prices; Sark reports rum 2.

Ship, PlayerShip and EnemyShip:

- Ship(...), attackStrength and isDefeated: a test subclass with maxHull 40 starts at hull 40; a fresh player attacks for 12 and is not defeated. maxHull 0 and a null crew throw IAE, a crew at morale 0 attacks for 7, and setHull(0) reports defeated.
- takeDamage, addGold, spendGold, addRum and takeRum: a fresh player taking 12 loses 10 hull, leaving hull 90, crew 18 and morale 60; 50 gold added or spent gives 250 or 150; 10 bottles added gives 40; takeRum(5) gives 5. Negatives throw IAE, takeDamage(500) returns 100 and leaves hull 0, spendGold(500) throws IAE with gold still 200, and takeRum(50) gives 30 and empties the hold.
- PlayerShip(...), moveTo and addNotoriety: a new ship sits at Sark with notoriety 0, sailing to the lane and back counts 1 port visited, and addNotoriety(2) gives 2. A null home, a move from Sark to Dover, and a negative notoriety each throw IAE.
- repair, hire and buyRum: from setHull(80), repair(10, 2) returns 10 leaving gold 180; hire(5, 15) returns 5; buyRum(20, 2) returns 20. repair(50, 2) at hull 80 returns 20 [13], hire(30, 15) returns 13 leaving gold 5 [12], and buyRum(20, 4) with 3 gold returns 0.
- upgradeCannons, upgradeArmour, payBonus and describe: an upgrade returns true at level 2 leaving gold 0; payBonus(100) after onWin(600) gives greed 25 and morale 82; describe holds every field. At level 5 or with 50 gold an

upgrade returns false [22], and payBonus(500) returns false with nothing changed.

- EnemyShip(...), describe and the EnemyType getters: (MERCHANT, 0) gives hull 40, crew 8, gold 150 and morale 30, (MERCHANT, 10) gives hull 60, crew 13, cannons 2 and gold 200, and MERCHANT.getGain() gives 1. (PIRATE, -1) and (null, 0) each throw IAE, and COAST_GUARD.getGain() gives 3.

Crew and Encounter:

- Crew(...), lose, hire, moraleFactor and isBroken: (20, 70) gives factor 0.85, not broken; lose(2) returns 2 leaving count 18 and morale 60; hire(5) returns 5. (50, 70) caps at 40, lose(25) returns 20 leaving morale 0 and a broken crew, hire(30) returns 20 [13], and negatives throw IAE.
- drink, onWin, onFlee and receiveBonus: with 30 bottles the crew drinks 2 and holds morale 70; onWin(150) gives morale 80 and greed 7; onFlee gives morale 55; a 100 gold bonus after onWin(600) gives greed 25 and morale 82. drink(1) drinks 1 and drops morale to 60, onWin(5000) caps greed at 100, and negatives throw IAE.
- Encounter(...), roll and describe: a sea lane with FixedRandom(0) gives a merchant encounter whose description holds "merchant". A null enemy throws IAE; a port, or the high seas with 99, gives null.
- fight and flee: a fresh player against (MERCHANT, 0) with roll 0 takes the surrender in round 3, ending at hull 97, gold 350, rum 38, morale 80 and greed 7; flee instead gives hull 92, crew 19 and morale 50. After setHull(5), the same fight against (PIRATE, 0) loses in round 1 with gold still 200, and flee empties the hull [21].

7. Division of work

Aidan owns CommandParser, Command, Ship, PlayerShip, EnemyShip, EnemyType, Crew and Encounter, with cases 1–6, 8–13, 18 and 21–27, and the final Web-CAT submission. Marcos owns Game, ChannelMap, Location, Port and Describable, with the map, all *default* numbers, cases 7, 14–17, 19 and 20, and the play-through transcript for the oral defence.

Day one, together: every class written with its section 4 signatures and stub bodies, so the project compiles and each of us edits only our own files. The week 1 milestone, at the Tuesday 7 pm meeting: start at Sark and run look, status, sail round all 18 stops, repair, hire, buy, help and quit. Week 2 adds fights, the crew rules, bonus, retire, mutiny and sinking, and we both play to set the numbers. Game is the only class calling both sides' code and Marcos owns it, so nobody edits the same file; each class lands on main with its test class, and the merged build must pass Web-CAT with no style warnings.

If time runs short we cut, each cut leaving every other class unchanged: greed and bonus first; then rum, with a fixed morale drop per sea move; then the three high-seas stops.

8. Revised scope

Each item gives the change, the reason, and where it came from.

1. An 18-stop metro map of ports, sea lanes and high seas, each kind with its own encounter chance, with sail moving one stop: the route chosen changes the risk and needs no route-finding. Marcos; group discussion dropped a 25-stop draft's coastal waters so a crossing takes two moves; port names from GenAI.
2. A HashMap from key to stop with an ArrayList of neighbours per stop, closing the \$5 unknown about coding the map. Aidan's \$5 idea and HashMap choice.
3. Crew morale, greed and rum moved from stretch goal 3 into the MVP in basic form (DESIGN.md FR15–17) with bonus added, so morale drives combat for both sides and fights gain a second ending. Marcos; the basic and full split from GenAI review.
4. An abstract Ship shared by both sides, with enemies scaled by EnemyType and notoriety: one fight routine, and the difficulty scaling promised in DESIGN.md §1 returns. Marcos asked for a shared base; the GenAI draft chose inheritance.
5. Cannon and armour levels on one price list in place of weapon items with per-port stock, removing an Item class and seven stock lists. GenAI review, after Marcos set a two-week budget.
6. Overflow (case 12) handled by dividing before multiplying, refusals as return values with dispatch writing every message, and a parse that never throws: no ArithmeticException or custom exception, and one place makes all player-facing text. Aidan's draft; the division from GenAI review.
7. The mode computed by Game.mode() with OVER as one value, closing the \$5 unknown about restricting commands by mode. Aidan's mode idea plus DESIGN.md's "state machine" note.
8. One Random injected through Game, with Encounter fully constructible, closing the \$5 unknown about forcing a fight in JUnit. GenAI review of \$5 in both drafts.
9. No enums: verbs, modes, kinds and nations are String constants with a HashMap verb table, and EnemyType is a class with three static final instances, because the course has covered HashMap, interfaces and generics but not enums. Marcos, after GenAI review proposed enums.